

llmXive Automated Review of Trust Region On-Policy Distillation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided

evidence (tables and text).

The primary concern lies in the **Introduction** (lines 68-70), where the authors state: “TrOPD substantially improves OPD by +3.34, +4.00, +5.11, and +6.18 points on math, code, instruction following, and STEM benchmarks, respectively.” A verification against the provided results tables reveals a discrepancy. In **Table 1** (lines 105-135), comparing TrOPD (49.85) to the OPD baseline (46.79) in the math domain yields a difference of +3.06, not +3.34. In **Table 2** (lines 330-360), the math domain improvement is +4.06 (52.08 vs 48.02). The specific values cited in the abstract and introduction do not correspond to the explicit row-by-row comparisons in the main results tables. This suggests either a calculation error in the text, a reference to a specific subset of data not clearly labeled, or a mismatch between the text and the final table values. This requires correction to ensure the claims accurately reflect the reported data.

Secondly, in **Section 3.1** (lines 165-167), the text claims that “naive reward clipping, as adopted by REOPOLD... may remove informative supervision... resulting in a performance bottleneck.” The data in **Table 1** shows REOPOLD (47.86) outperforming the vanilla OPD baseline (46.79). While TrOPD (49.85) is superior, characterizing REOPOLD’s result as a “performance bottleneck” relative to the baseline is factually inconsistent with the table, which shows an improvement. The phrasing should be adjusted to clarify that while REOPOLD improves upon the baseline, it hits a ceiling that TrOPD surpasses, rather than implying it causes a bottleneck compared to the unmodified OPD.

Finally, the claim regarding **Figures 3 and 4** (lines 230-232) asserts that TrOPD achieves a lower gradient norm than both OPD and Clip Outlier. Since the rendered figures are not visible, this claim cannot be fully verified. However, the text explicitly contrasts TrOPD with Clip Outlier. The authors must ensure the visual data in the figures explicitly supports the specific comparison that TrOPD’s gradient norm is lower than Clip Outlier’s, as the text implies a direct visual confirmation of this specific relationship.

The rest of the claims regarding the methodology (TrOPD formulation, trust region definition) appear consistent with the equations and the ablation studies presented in Table 3. The citation of REOPOLD and other baselines appears consistent with the provided bibliography.

Action items.

- **[writing]** Introduction claims TrOPD improves OPD by +3.34, +4.00, +5.11, +6.18 points. Table 1 shows math gain is +3.06 (38.54 vs 35.83). Table 2 shows +4.06. The text numbers do not match table data. Correct the values or specify the exact config used.
- **[writing]** Section 3.1 claims REOPLD causes a ‘performance bottleneck’ by removing supervision. Table 1 shows REOPLD (47.86) outperforms OPD baseline (46.79). The claim contradicts the data showing improvement. Rephrase to state REOPLD improves baseline but is outperformed by TrOPD.
- **[writing]** Section 3.2 claims TrOPD has lower gradient norm than Clip Outlier per Figures 3/4. Figures are not visible. Ensure the figures explicitly show $\text{TrOPD} < \text{Clip Outlier}$ for gradient norms, not just general trends, to support this specific comparison.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure presents clear performance data across four benchmarks, but the caption contains a formatting error with an embedded citation and lacks clarity on the specific baseline model used for the ‘Qwen3-1.7B’ comparison.

Figure 2

Figure 2 provides a conceptual overview of the distillation process but relies on unexplained algebraic analogies and contains contradictory labeling in the ‘Off-Policy Guidance’ section.

Figure 3

The figure presents a clear line plot of entropy over steps with a legend, but the caption is insufficiently descriptive to explain the specific methods (‘OPD’, ‘Clip Outlier’, ‘Mask Outlier’) being compared or the significance of the observed trends.

Action items.

- **[science]** Figure 1: The caption states that TrOPD, OPD, and REOPOLD are trained on ‘Qwen3-SFT-1.7B’, but the x-axis labels include ‘Qwen3-1.7B’ as a distinct baseline. The caption fails to clarify if this baseline is the raw SFT model or the Base model, creating ambiguity regarding the comparison group.
- **[writing]** Figure 1: The citation ‘ko2026scaling’ is embedded directly into the figure caption text (‘...REOPOLD ko2026scaling...’) rather than being formatted as a standard reference, which disrupts readability.
- **[science]** Figure 2: The diagram uses specific algebraic equations (e.g., $(x-2)(x-3)=0$) to illustrate the ‘Trust Region’ and ‘Outlier’ concepts, but the caption and visual labels do not explain the mathematical analogy or how these equations map to the probability distributions shown on the left.
- **[science]** Figure 2: The ‘Off-Policy Guidance’ section displays a ‘Fully On-Policy from Student’ example, which appears contradictory to the section header and the concept of off-policy guidance.
- **[writing]** Figure 2: The text ‘K1 RKD’ and ‘Topk FKD’ appears in the diagram without

definition in the caption or visual legend, making the specific method names unclear.

- **[science]** Figure 3: The legend lists ‘OPD’ (dashed) and ‘Clip Outlier’ (dotted), but the caption ‘Entropy comparison’ is too vague to explain why these specific methods are being compared or what the ‘Outlier’ variants represent in this context.
- **[writing]** Figure 3: The legend entry ‘OPD’ corresponds to a dashed line, but the plot shows a dashed line that drops significantly lower than the others; without a clear definition of what ‘OPD’ refers to in this specific entropy context (e.g., baseline vs. proposed), the comparison is ambiguous.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and shorthand that are not defined upon first use, creating barriers for readers outside the immediate sub-field of on-policy distillation.

In Section 3.1 (“Distillation for Language Models”), the authors introduce “teacher of generations (ToGs)” and “student of generations (SoGs)” but immediately switch to using the acronyms “ToG” and “SoG” without explicitly defining the acronym form (e.g., “ToG” vs “ToGs”). While the context is clear to an expert, a general reader may stumble. Similarly, in the Abstract and Section 5.1, the term “SoTA” is used repeatedly. This should be expanded to “state-of-the-art” at the first instance.

Furthermore, the experimental section (Section 5.2) is dense with benchmark acronyms—AIME, AMC, GPQA, MMLU, IFBench, LCB—none of which are expanded in the text. While these are standard in the LLM community, a paper claiming broad applicability should define them (e.g., “American Invitational Mathematics Examination (AIME)”) to ensure accessibility.

Finally, the “Limitations” section mentions “mid-training” as a necessary stage. This is a non-standard term in the broader machine learning literature and is used without definition, potentially confusing readers regarding the specific training phase being referenced.

These issues are minor in terms of scientific validity but significant for readability and inclusivity. The authors should perform a pass to expand all acronyms at their first occurrence and define non-standard terminology.

Action items.

- **[writing]** Define ‘SoG’ (Student of Generation) and ‘ToG’ (Teacher of Generation) at first use in Section 3.1. The acronyms are introduced without definition, relying on reader inference from context.
- **[writing]** Replace ‘SoTA’ with ‘state-of-the-art’ in the Abstract and Section 5.1. Acronyms for common phrases should be spelled out on first occurrence to aid non-specialist readers.

- **[writing]** Define ‘AIME’, ‘AMC’, ‘GPQA’, ‘MMLU’, and ‘IFBench’ at their first mention in Section 5.2 or the Introduction. Currently, these benchmark acronyms are used without expansion, assuming domain familiarity.
- **[writing]** Clarify the term ‘mid-training’ in the Limitations section. This is not a standard, universally defined term in the field and requires a brief explanation of what stage of the pipeline it refers to.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical consistency of the paper is generally sound in its high-level motivation: identifying that distribution mismatch causes unstable gradients in On-Policy Distillation (OPD) and proposing a trust-region mechanism to mitigate this. However, there are specific logical gaps in the mathematical justification and data reporting that undermine the internal consistency of the arguments.

First, in Section 3.2 under “Outlier Estimation,” the paper argues that the auxiliary Forward KL (FKL) objective in outlier regions is “suppressed” when the student probability mass on the teacher’s top-k tokens approaches zero. The text states: “when $\sum_{v \in \mathcal{V}_T^k} \pi_S(v) \rightarrow 0$, we have $\text{KL}(\pi_T \parallel \pi_S) \rightarrow 0$.” This is a logical error. The definition of KL divergence is $\sum \pi_T \log(\pi_T/\pi_S)$. If $\pi_S(v) \rightarrow 0$ for any v where $\pi_T(v) > 0$, the term $\log(\pi_T/\pi_S) \rightarrow \infty$, causing the KL divergence to explode, not vanish. The authors likely intended to say the *gradient* is masked or the term is weighted by the mask $\overline{\mathbb{M}}_x$, but the stated mathematical limit contradicts the definition of the divergence used. This breaks the causal link between the proposed mechanism and the claimed stability.

Second, the definition of the adaptive trust region in Section 3.2 introduces a stochastic mask $\mathbb{M}_x \sim \text{Bernoulli}(P_{\text{trust}}(x))$ where $P_{\text{trust}}(x) = \min(\pi_T(x)/\pi_S(x), 1)$. While this mimics speculative decoding acceptance, the paper does not explicitly derive how the expectation of the gradient over this Bernoulli distribution relates to the final objective function. Specifically, if the mask is applied stochastically during training, the gradient estimator must account for the probability of the mask being 1. The unified objective equation (Eq. 10) presents the terms as deterministic products of indicators and losses, which is inconsistent with the stochastic definition of $\mathbb{M}_x$ unless the expectation is taken inside the loss calculation in a way not fully detailed. This creates a gap between the proposed algorithm and the stated optimization objective.

Finally, there is a numerical inconsistency in the results reporting. In Section 4.3 (“Ablation Studies”), the text claims: “the three TrOPD variants... outperform OPD by 2.00, 1.94, and 3.06 points on average, respectively.” The “3.06 points” figure is attributed to the “TrOPD FKL” variant. However, Table 4 (Ablation Studies) shows OPD averaging 46.79 and TrOPD FKL averaging 49.85. The difference is $49.85 - 46.79 = 3.06$. Wait, the text says “outperform OPD

by... 3.06 points”. Let’s re-read the table. OPD is 46.79. TrOPD FKL is 49.85. The difference is indeed 3.06. However, the text also says “TrOPD outperforms it [REOPOLD] by 1.99 and 1.84 points”. In Table 3, REOPOLD is 38.79 and TrOPD is 40.63. The difference is 1.84. The text says 1.99 for math and 1.84 for general. This seems consistent. Let’s re-examine the “FKL Outlier” comparison. The text says “TrOPD outperforms... FKL Outlier... by 3.06 points”. Table 4 shows FKL Outlier at 49.00 and TrOPD FKL at 49.85. The difference is 0.85. The text claims 3.06. This is a direct contradiction between the text’s claim of the magnitude of improvement over the FKL Outlier baseline and the data presented in the table. The text likely confused the improvement over the *vanilla OPD* (which is 3.06) with the improvement over the *FKL Outlier* baseline. This misrepresentation of the ablation results weakens the logical support for the claim that the full TrOPD method provides a significant gain over the specific “FKL Outlier” component.

Action items.

- **[science]** In Section 3.2, the paper claims that when $\sum(\pi_S(v)) \rightarrow 0$, $\text{KL}(\pi_T \parallel \pi_S) \rightarrow 0$. This is mathematically incorrect; if $\pi_S(v)$ approaches 0 while $\pi_T(v) > 0$, the term $\log(\pi_T/\pi_S)$ approaches infinity, causing KL to diverge. The logic for why the outlier objective is ‘suppressed’ needs correction or a different mechanism explanation.
- **[science]** The definition of the trust region probability $P_{\text{trust}}(x) = \min(\pi_T(x)/\pi_S(x), 1)$ is presented as a Bernoulli probability. However, if $\pi_S(x)$ is very small, this ratio can exceed 1, which is clamped to 1. The paper does not explain how this stochastic masking interacts with the gradient estimator for the RKL term, specifically whether the expectation over the Bernoulli mask is correctly accounted for in the final objective function.
- **[writing]** Table 1 lists ‘FKL Outlier’ with an average score of 49.00, while ‘TrOPD’ (which includes the same FKL Outlier component plus off-policy guidance) scores 49.85. However, the text in Section 4.3 claims TrOPD outperforms ‘FKL Outlier’ by 3.06 points on average. The numbers in the text ($49.85 - 49.00 = 0.85$) contradict the claimed 3.06 point gain, suggesting a calculation error or a mismatch in the reported baselines.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper exhibits over-claiming in three primary areas regarding the specificity of results, the definition of contributions, and the mechanistic attribution of success.

First, the Abstract and Introduction assert that TrOPD achieves specific point improvements: “+3.34, +4.00, +5.11, and +6.18 points on math, code, instruction following, and STEM benchmarks.” However, the provided tables (Table 1, Table 2, Table 3) do not explicitly display these exact aggregate numbers. For instance, Table 2 reports an average improvement of +3.06

on math and +2.63 on general domains, which conflicts with the specific breakdown in the abstract. Without a transparent derivation or a table showing these exact per-domain deltas, the claim of these precise figures constitutes an over-claim of precision and potentially misrepresents the data’s granularity.

Second, the Contributions section states, “We establish a general benchmark for reasoning-oriented OPD.” The manuscript describes evaluating existing methods (OPD, EOPD, RE-OPOLD) on existing datasets (AIME, LiveCodeBench, etc.) under unified settings. While this constitutes a valuable “unified evaluation” or “comparative study,” it does not meet the standard definition of “establishing a benchmark,” which typically implies creating a new dataset, metric, or protocol. This phrasing overstates the novelty of the experimental setup.

Finally, the claim that the method specifically “mitigates the optimization difficulty of the K1 reverse-KL estimator” (Abstract) is a strong causal assertion. While the paper demonstrates improved stability and performance, the evidence provided (primarily final accuracy scores and a referenced gradient norm figure) does not explicitly isolate the K1 estimator’s instability as the sole or primary failure mode being addressed, distinct from general distribution mismatch issues. The paper would benefit from more direct evidence, such as training curves showing collapse in baselines specifically due to K1 variance, to support this specific mechanistic claim.

Action items.

- **[writing]** Abstract claims specific gains (+3.34, +4.00, etc.) not explicitly derived in tables. Tables show different averages (e.g., +3.06 math). Clarify or remove precise numbers to avoid over-claiming precision.
- **[writing]** Claim to ‘establish a general benchmark’ overstates contribution. The work performs a unified evaluation on existing datasets, not creating a new benchmark. Rephrase to ‘conduct a unified evaluation’.
- **[science]** Claim that TrOPD specifically mitigates ‘K1 reverse-KL estimator’ instability lacks direct evidence isolating K1 failure modes. Provide training curves or analysis linking stability gains specifically to K1 mitigation.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript focuses on algorithmic improvements for on-policy distillation (TrOPD) and does not involve human subjects, animal testing, or direct collection of private data, thus avoiding standard IRB/IACUC requirements. However, several safety and ethics considerations regarding data provenance and potential misuse warrant clarification.

First, the Appendix details the training data for the teacher model Qwen3-Nemotron-4B, which includes “WildChat-1M” and “Open-Instruct” (Appendix, lines 340-345). While these

are public datasets, the distillation process transfers knowledge from the teacher to the student. The authors should explicitly confirm that these source datasets were vetted for personally identifiable information (PII) and that the distillation process does not amplify the memorization of sensitive user data present in the teacher’s training corpus. A brief statement on data privacy compliance regarding the source datasets would strengthen the ethical standing of the work.

Second, the “Off-Policy Guidance” mechanism (Section 3.3) utilizes teacher-generated trajectories to guide the student. If the teacher model was trained on data containing copyrighted code or sensitive proprietary information, there is a risk that the student model could inherit and reproduce these patterns. The authors should briefly discuss whether any filtering was applied to the teacher’s output data to prevent the propagation of potentially copyrighted or sensitive content into the distilled student model.

Finally, while the benchmarks (math, code, STEM) are standard, the resulting models are “Small Reasoning Models” capable of complex logic. The paper should acknowledge the dual-use potential of such models, particularly in code generation, where they could be used to generate malicious scripts or exploit vulnerabilities. A sentence in the Limitations or Conclusion addressing the responsible deployment of these models and the need for safety alignment in downstream applications would be appropriate.

Overall, the paper does not present immediate fatal ethical flaws, but these clarifications are necessary to ensure full transparency regarding data privacy and potential misuse.

Action items.

- **[writing]** The paper relies on teacher models (e.g., Skywork-OR1-Math-7B, Qwen3-Nemotron-4B) trained on datasets including WildChat-1M and Open-Instruct (Appendix). Explicitly state whether these datasets contain personally identifiable information (PII) and confirm that appropriate consent or public-domain status was verified for the distillation process to ensure data privacy compliance.
- **[writing]** The ‘Off-Policy Guidance’ section describes using teacher-generated prefixes to guide student generation. Clarify if the teacher model’s training data includes copyrighted material or sensitive content that could be inadvertently memorized and regurgitated by the student, and discuss any mitigation strategies for potential IP or safety leakage.
- **[writing]** The methodology involves training on mathematical reasoning and code generation benchmarks. While generally benign, acknowledge potential dual-use risks if the distilled models are applied to generate malicious code or exploit mathematical vulnerabilities, and briefly mention safety alignment considerations in the Limitations or Conclusion.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the claims of TrOPD is generally strong in terms of experimental breadth, covering multiple domains (math, code, STEM) and teacher-student configurations. However, the statistical rigor regarding the reported performance gains requires clarification.

First, while the “Benchmark Evaluation” section (lines 338-342) states that mathematical reasoning results are averages of 32 evaluation runs, the results tables (Table 1, Table 2, Table 3) report only point estimates. There is no mention of standard deviations, confidence intervals, or statistical significance tests (e.g., paired t-tests) to confirm that the observed improvements (e.g., +3.06 points in single-domain distillation) are not due to random variance. Given the stochastic nature of LLM generation and policy gradient updates, reporting variance is essential to validate the robustness of the proposed method.

Second, the ablation studies in Table 3 (lines 305-322) present a compelling breakdown of the TrOPD components. However, the comparison between “FKL Outlier” and “TrOPD FKL” (which adds off-policy guidance) relies on comparing average scores. The paper does not explicitly state whether these specific ablation runs were conducted with the same random seeds or if the differences are statistically significant. A more rigorous ablation would pair the runs or report the variance across multiple seeds for each ablation variant.

Third, the core mechanism of the “Adaptive Trust Region” relies on the probability $P_{trust}(x) = \min(\pi_T(x)/\pi_S(x), 1)$ (Equation 8, line 228). The paper asserts that this effectively partitions tokens into reliable and unreliable regions but provides no empirical data on the distribution of these probabilities during training. A histogram showing the fraction of tokens falling into the trust region versus the outlier region would strengthen the evidence that the method is dynamically adapting as the student policy improves, rather than simply masking a static portion of the data.

Finally, the memory complexity claims in Table 2 (line 256) are theoretical. While the $\mathcal{O}(nk)$ complexity for the outlier FKL term is noted, the paper does not provide empirical wall-clock time or GPU memory usage comparisons against the baselines (OPD, REOPOLD). Given that the outlier estimation involves a top- k summation, the practical overhead could be significant. Including a brief analysis of the computational cost would provide a more complete picture of the trade-offs involved.

Action items.

- **[science]** The paper claims TrOPD outperforms baselines by specific margins (e.g., +3.34 on math, Table 1 caption) but does not report standard deviations or statistical significance tests (e.g., t-tests) across the 32 evaluation runs mentioned in ‘Benchmark Evaluation’. Without variance estimates, the robustness of these gains against random seed noise is unverified.
- **[science]** The ablation study in Table 3 (tab:ablation) compares TrOPD variants but lacks a direct comparison against a ‘Vanilla OPD’ baseline trained under the exact same random seeds and hyperparameters for the specific ablation setting. The claim that FKL Outlier is superior relies on comparing results across different table rows which may not be statistically paired.
- **[science]** The ‘Adaptive Trust Region’ probability $P_{trust}(x)$ is defined as $\min(\pi_T(x)/\pi_S(x), 1)$. The paper does not provide empirical evidence (e.g., a histogram or distribution plot) showing

the actual fraction of tokens classified as ‘trust region’ vs. ‘outlier’ during training. Without this, the claim that the method effectively partitions the space remains theoretical.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis presented in the manuscript lacks necessary rigor to support the claimed improvements with high confidence. While the experimental design covers multiple domains and baselines, the reporting of results is insufficient for statistical validation.

Specifically, the “Benchmark Evaluation” section states that mathematical reasoning results are “the average accuracy of 32 times evaluation.” However, none of the result tables (Table 1, Table 2, Table 3) report the standard deviation, standard error, or confidence intervals associated with these averages. In reinforcement learning and distillation tasks, performance can vary significantly due to sampling stochasticity. Without measures of variance, it is impossible to determine if the reported gains (e.g., the +3.06 point improvement over OPD in single-domain distillation) are statistically significant or within the margin of error. The authors should re-run the analysis to include standard deviations and perform appropriate significance testing (e.g., paired t-tests or bootstrap confidence intervals) against the baselines.

Furthermore, the proposed “Adaptive Trust Region” mechanism (Eq. 10) introduces a stochastic mask $\mathbb{M}_x \sim \text{Bernoulli}(P_{\text{trust}}(x))$. The manuscript does not discuss the variance of the gradient estimator resulting from this stochastic masking. In policy gradient methods, high variance in the estimator can lead to unstable training or biased updates. The authors should clarify if variance reduction techniques (such as control variates) are employed or provide empirical evidence (e.g., gradient norm stability plots with error bands) that the stochastic masking does not introduce detrimental noise.

Finally, the aggregation method for the “Avg.” columns in the tables is ambiguous. It is unclear if these represent the mean of the benchmark averages or the mean of all individual task accuracies. For reproducibility and accurate meta-analysis, the exact aggregation formula and the underlying variance metrics must be explicitly stated.

Action items.

- **[science]** The paper claims results are ‘average accuracy of 32 times evaluation’ (Benchmark Evaluation) but does not report standard deviation, standard error, or confidence intervals for any metric in Tables 1, 2, or 3. Given the stochastic nature of LLM sampling and the small number of seeds ($n=32$), statistical significance testing (e.g., paired t-tests or bootstrap CIs) is required to validate that observed gains (e.g., +3.06 points) are not due to random variance.
- **[science]** The adaptive trust region probability $P_{\text{trust}}(x)$ is defined as a Bernoulli trial (Eq.

10). The manuscript does not specify the number of samples used to estimate the expectation of this stochastic mask or how the variance introduced by this sampling is handled during backpropagation. Clarification on the variance reduction technique (e.g., control variates) or the stability of the gradient estimator is needed.

- **[science]** Table 1 and Table 2 report single-point performance metrics without error bars. For the ‘Avg.’ columns, the aggregation method (mean of means vs. mean of all samples) is not explicitly defined. Reproducibility requires reporting the standard deviation across the 32 evaluation runs for each benchmark to assess the robustness of the reported improvements.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a generally high level of writing quality, with clear exposition of complex technical concepts and a logical flow throughout the paper. The abstract effectively summarizes the problem, method, and results. However, there are several minor grammatical and stylistic issues that, while not severely impeding understanding, detract from the overall polish and professionalism of the text.

Specifically, in Section 3.2 (“Adaptive Trust Region”), the sentence “Different from previous predefined threshold , we define the trust region...” contains a grammatical error. The phrase “previous predefined threshold” should be pluralized to “thresholds” or rephrased as “Unlike the previous predefined threshold ” to correctly refer to the singular concept or the general class of methods. This error slightly disrupts the reading flow.

In Section 4.1 (“Benchmark Evaluation”), the phrase “each result is the average accuracy of 32 times evaluation” is awkward and non-idiomatic. It should be revised to “each result is the average accuracy over 32 evaluation runs” or “evaluated 32 times” to sound more natural and precise.

Additionally, in the Introduction, the sentence introducing the three perspectives of the benchmark (“This work establishes a unified benchmark to systematically study this challenge from three perspectives: (1)...”) could be slightly improved for flow. While the list is clear, the transition from the colon to the numbered items feels a bit abrupt. Rephrasing to “through three perspectives” or integrating the list more smoothly into the sentence structure would enhance readability.

Finally, in Section 3.1, the phrase “stand-alone FKL can not achieve effective training” uses “can not” which is often considered less formal than “cannot” in academic writing. Furthermore, the sentence structure is somewhat passive; changing it to “stand-alone FKL fails to achieve effective training” would be more direct and impactful.

Addressing these minor issues will further improve the clarity and professionalism of the manuscript. The core scientific narrative is well-presented, and these edits are purely stylistic

and grammatical refinements.

Action items.

- **[writing]** In Section 3.2, the sentence ‘Different from previous predefined threshold , we define the trust region...’ contains a grammatical error. It should read ‘Different from previous predefined thresholds ’ or ‘Unlike the previous predefined threshold ’ to ensure subject-verb agreement and clarity.
- **[writing]** In Section 4.1, the phrase ‘32 times evaluation’ is awkward and non-idiomatic. It should be revised to ‘evaluated 32 times’ or ‘based on 32 evaluation runs’ for better readability and professional tone.
- **[writing]** In the Introduction, the sentence ‘This work establishes a unified benchmark to systematically study this challenge from three perspectives: (1)... (2)... and (3)...’ uses a colon followed by a list that is not fully integrated into the sentence structure. Consider rephrasing to ‘This work establishes a unified benchmark to systematically study this challenge through three perspectives: (1)...’ or restructuring the list to flow more naturally.
- **[writing]** In Section 3.1, the phrase ‘stand-alone FKL can not achieve effective training’ uses ‘can not’ which is often considered less formal than ‘cannot’. Additionally, the sentence structure is slightly clunky; consider ‘stand-alone FKL fails to achieve effective training’ for conciseness and impact.